



Pharm*Achieve*

BURNS AND SUNBURNS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

5-FU 5-Fluorouracil

ASA Acetylsalicylic Acid

BCC Basal Cell Carcinoma

BID Twice Daily

BSA Body Surface Area

DEET N,N-Diethyl-M-Toluamide

EGF Epidermal Growth Factor

HCTZ Hydrochlorothiazide

IM Intramuscular

IV Intravenous

NSAID Non-Steroidal Anti-Inflammatory Drug

PABA Para-Aminobenzoic Acid

PO Per Os (Oral)

PRN As Needed

Q Every

SC Subcutaneous

SCC Squamous Cell Carcinoma

SPF Sun Protection Factor

TBSA Total Body Surface Area

UV Ultraviolet

UVA Ultraviolet A

UVB Ultraviolet B

UVC Ultraviolet C

BURNS

Qualifying Exam Preparatory Course



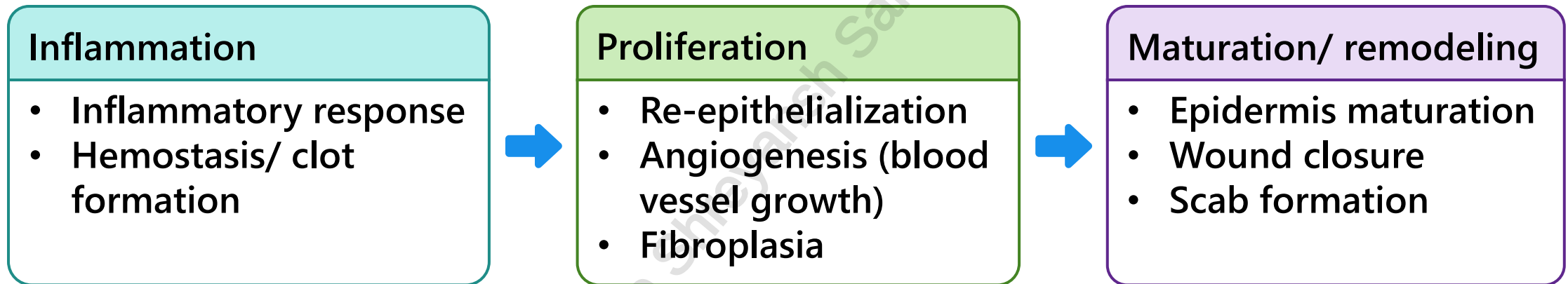
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OBJECTIVES

- 1 Define and classify burns
- 2 Discuss the pathophysiology, risk factors, and clinical presentation
- 3 Determine when self-treatment is appropriate
- 4 Discuss treatment and management of burns

DEFINITIONS AND PATHOPHYSIOLOGY

- A burn is an acute injury characterized by the destruction of cells to the skin or other tissue
- The healing process depends on the severity of the burn
- Burns are categorized by: burning agent, depth, % total Body Surface Area (BSA)
- Wound healing is divided into 3 main steps:



TYPES OF BURNS

Thermal burns

- Most common type of burn
- Contact: burn as a result of hot objects
- Flame: burn as a result of an open flame
- Scald: burn as a result of contact with hot liquids or steam

Cold thermal

- Also known as frostbite
- Ice crystals puncture cells, leading to hypoxia

TYPES OF BURNS

Electrical

- Current passes through tissue

Chemical

- Altered pH disrupts cell membrane
- Can occur via absorption, inhalation or ingestion

CLASSIFICATION OF BURNS

Burns are classified by the depth of the injury:

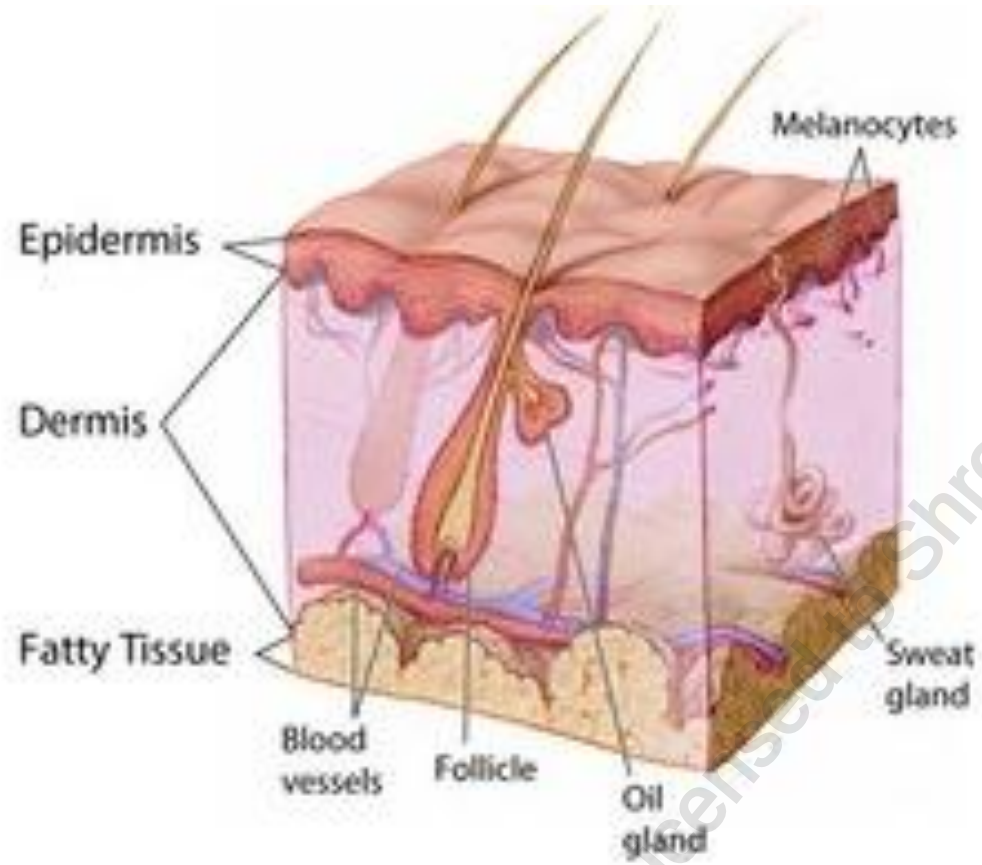


Image courtesy of Don Bliss. Skin Anatomy. National Cancer Institute. May 2010.

CLASSIFICATION OF BURNS

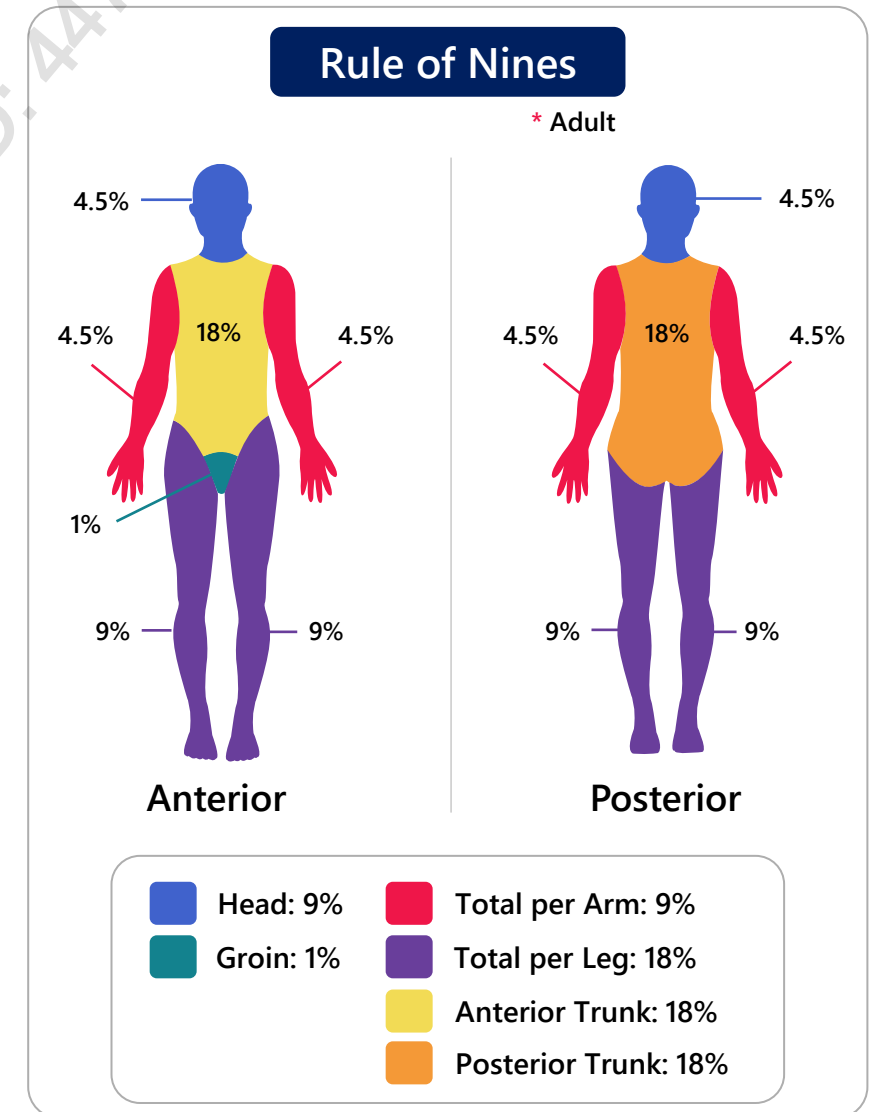
	Superficial	Superficial partial thickness	Deep partial thickness	Full thickness	Full thickness
Degree	1 st degree	2 nd degree	2 nd degree	3 rd degree	4 th degree
Clinical Presentation	- Painful - Dry - Erythema - No blisters - Blanches upon pressure - Involves epidermis	- Extremely painful - Moist - Erythema - Blanches upon pressure - Weeping (fluid leaking) - Blisters - Involves dermis	- Pain upon pressure - Blisters - Easily broken - Wet or waxy dry - Patchy white or red - No blanching with pressure - Involves epidermis, upper and deep dermis - Damages hair follicles and glands - Sweat and hair follicle damage	- May be painless as a result of nerve damage - Dry - White, grey, or black skin - Waxy or leathery appearance - No blisters - No blanching with pressure - Involves epidermis, dermis, subcutaneous tissue	- Life-threatening - Extends through skin into fascia, muscle or bone
Duration	3-6 days	1-3 weeks	3-9 weeks	Requires surgery +/- grafts No natural healing	Requires surgery +/- grafts No natural healing
Scarring	No scarring	Discoloration of skin No long-term scarring	Risk of scarring +/- contracture (skin tightening)	High risk of scarring and contracture	High risk of scarring and contracture

EXTENT OF THE BURN

- The size of the burn is assessed and compared to the total body surface area (TBSA)
- The percentage of affected TBSA is used to assess only partial thickness or full thickness burns
- % of TBSA is used to determine treatment or referral to burn center

EXTENT OF THE BURN

- Two quick methods to estimate % TBSA are:
 - Rule of Nines (Adults):
 - Head equals 9% TBSA
 - Each arm equals 9 % TBSA
 - Each leg equals 18 % TBSA
 - Each trunk (anterior and posterior trunk) equals 18% TBSA
 - Groin equals 1 % TBSA
 - Palm Method:
 - Palm with fingers equals 1% TBSA
- Lund-Browder Rule (Children):
 - Considers age of person
 - Burn area is transcribed onto a chart
 - Body Surface Area (BSA) calculated based on chart rules



RISK FACTORS

Age: children or elderly

- <5 or >60 years of age

Low-income countries

- Inadequate safety procedures and measures

Socioeconomic status

- Unemployment, poor education, crowded living conditions, malnutrition

Occupation

- Greater exposure to causes increase the risk of a burn

Drug abuse, smoking

Medical conditions

- Epilepsy, cognitive disabilities, diabetes, vascular disease

CLINICAL PRESENTATION

Consider the following when assessing the severity of a burn:

- **Location:** thin skin vs thick skin area
- **Type of burn:** thermal, electrical, chemical, radiation
- **Depth of burn:** superficial, partial thickness, full thickness
- **TBSA of burn:** % of total body surface area affected

WHEN TO REFER TO EMERGENCY

- Burns covering >15% BSA in adults and >10% TBSA in children
- Any burn that is more serious than a mild, superficial partial thickness burn
- Burns involving thin skin areas (e.g. face, eyes, ears, perineum)
- Burns involving hands, feet, or circumferential areas
- Burns in children <5 years and adults >60 years (thin skin)
- Burns involving concomitant trauma
- Electrical and chemical burns, even if minor due to risk of progression to deep tissue necrosis within 24 hours



 Consider referral and close monitoring in patients with underlying conditions like diabetes, cirrhosis, or those who are immunocompromised

GOALS OF THERAPY

- ✓ Prevent or minimize skin damage
- ✓ Promote healing
- ✓ Minimize loss of function
- ✓ Reduce morbidity and mortality
- ✓ Prevent complications and secondary infection
- ✓ Provide symptom relief, wound-management and follow-up

NON-PHARMACOLOGICAL THERAPY

Mild Burn	Moderate - Severe Burn
Separate victim from afflicting agent	
Remove any debris, clothing, and jewelry from the burned area	Avoid this step in major burns
Reduce pain and limit inflammation by running cool water over the affected area for 20 minutes	Avoid this step in large burns due to risk of hypothermia
Gently cleanse the burn wound with mild soap and water; do NOT use disinfectants	
Apply cool sterile compress to the affected area	
Keep the wound clean and covered if the skin is broken	
Do NOT use ice as it may cause vasoconstriction and lead to frostbite	
Do NOT use home remedies, such as butter or grease, as it traps in the heat which worsens the burn and increases the risk of a secondary infection	

NON-PHARMACOLOGICAL THERAPY

- Dressings are placed over the burn area to:
 - Provide barrier against contamination
 - Absorb drainage
 - Maintain moisture
- Open therapy (no dressing) is typically used for burns on the head, neck, and groin
- Dressing changes vary:
 - Dressings must adequately control wound exudate
 - More exudate = more frequent dressing changes
 - Dressings should not be changed too frequently
 - May interfere with healing process
 - Dressings should be changed if the patient notices: change in smell, if it is wet/soggy, painful or dirty

NON-PHARMACOLOGICAL MEASURES: COMPRESS DRESSINGS

Fine mesh gauze	• Commonly contains antimicrobial properties; most common compress dressing
Hydrocolloid dressings	• Contains sodium carboxymethylcellulose, gelatin, and pectin • Inner gel layer absorbs exudate; autolytic debridement
Hydrogel dressings	• Rehydrate wound bed; autolytic debridement; for dry/low to moderate exuding wounds
Silver-containing dressings	• Activated silver has antimicrobial effects • Staining of the wound and surrounding skin along with stinging and sensitization can occur • Discontinue after 2 weeks if no improvement and re-evaluate • Useful for low to high exuding wounds or if clinical signs of infection
Polyurethane film	• Transparent and prevent bacterial contamination • Should not be used in infected or heavily exuding burns

NON-PHARMACOLOGICAL MEASURES FOR SELF-TREATABLE MINOR BURNS

Pain control

- Acetaminophen or NSAIDs; avoid ASA due to bleeding risk
- Limited evidence for topical anesthetics (lidocaine, benzocaine, pramoxine) or counterirritants (camphor, menthol, phenol)
- Opioids often time prescribed for deeper burns

Pruritus control

- Use moisturizing lotions and colloidal oatmeal baths; consider systemic antihistamines
- Do not use topical antihistamines due to risk of sensitization and allergic contact dermatitis

Antibacterial agents

- Topical antiseptics and antibiotics are not routinely recommended in minor burns
- Significant debate regarding efficacy of routine use of topical antibiotics
- Avoid silver sulfadiazine due to poorer healing outcomes
- Systemic antibiotics should not be used to prevent infections

NON-PHARMACOLOGICAL MEASURES FOR SELF-TREATABLE MINOR BURNS

Sunscreen

- Use SPF30+ sunscreen to prevent hyperpigmentation

Skin protectants

- Petrolatum in open areas for moisture retention and barrier to contamination
- Limited evidence for honey and aloe vera

PHARMACOLOGICAL THERAPY

Patients who require referral should undergo emergency care:

Assessment	Clinical presentation	Management
Smoke Inhalation	Cough, stridor, wheezing, neck burns, respiratory distress, hypoxia, oropharynx edema, elevated carbon monoxide	• Supplemental oxygen • Intubation
Carbon Monoxide Poisoning	Headache, nausea, confusion, dizziness	• Check carboxyhemoglobin levels • Hyperbaric oxygen treatment • Hydroxocobalamin (DO NOT USE THIOSULFATE)
Tetanus	Prophylaxis: bacteria that produce the tetanus toxin can infect burn victims or broken skin	• Check tetanus immunization • Administer tetanus prophylaxis

PHARMACOLOGICAL THERAPY: MODERATE-SEVERE BURNS

Topical antibiotics

- Routinely used for initial management, but efficacy is debatable
- Remove any debris, ruptured blisters, and loose tissue from the burn (debridement) then clean the affected area
- Apply topical antibiotics (2 mm thick BID) to all deep partial-thickness and full thickness burns
- Apply a dressing over the topical antibiotic OR apply a dressing that contains antibiotics
- Equivalent efficacy among topical antibiotics, choice based on cost and ease of use

PHARMACOLOGICAL THERAPY

Drug and antibacterial activity	Comments
Bacitracin; Gram Positive Cocci/Bacilli	• Commonly in combination with other active ingredients • Transparent, easy to apply, cosmetically acceptable • Poor eschar penetration; risk of contact dermatitis • Moderate antibacterial spectrum
Framycetin 1% Gram Positive Cocci, Gram Negative Bacilli	• Easy application • Poor penetration, moderate antibacterial action
Fusidic acid 2%; Gram Positive: <i>S. Aureus</i> , <i>Streptococcus spp.</i> , <i>Corynebacterium spp.</i> , <i>Clostridia spp.</i> Gram-negative: <i>Neisseria spp.</i> , <i>Bacteroides spp</i>	• May develop resistance • Mainly active against gram-positive organisms
Mupirocin 2%; <i>S. Aureus</i> , <i>Streptococci</i>	• Limited coverage against gram negative bacteria
Silver sulfadiazine 1%; Broad spectrum	• Only fair penetration; avoid in pregnancy and breastfeeding • Limited applicability, multiple dressing changes, poorer wound healing than other dressings, questionable efficacy; avoid in sulfa allergy
Chlorhexidine $\leq 2\%$; Broad spectrum	• Bacteriostatic (low conc.) and bacteriocidal (high conc.) • Concern for wound healing at higher concentrations ($>2\%$) • Avoid use on face and scalp due to toxicity
Povidone/Iodine 1%; Broad spectrum	• Poor penetration, tissue staining, painful • Avoid in pregnancy and breastfeeding due to possible iodine toxicity

PHARMACOLOGICAL THERAPY: MODERATE-SEVERE BURNS

Analgesics

- NSAIDs (ibuprofen 400 mg Q4-6H)
- Oral/IV opioid analgesics: morphine, hydromorphone, fentanyl can be used for severe pain (avoid IM or SC)

Surgery or skin grafts

- Used to treat full thickness burns

TIPS

- Self-care is **ONLY** for minor burns and sunburns
 - Refer to physician if healing is incomplete after 7 days
 - Refer to physician if more than mild burn or those with severe pain
- Monitor for signs of infection
 - Swollen lymph nodes
 - Fever
 - Pus
 - Increase in redness and swelling

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Frequency of monitoring/Timeframe
Burn	Healed	Daily for 2-3 days, then weekly until complete healing
Pruritus	Improve/alleviate	Every 4-6 hours for the first few days then daily thereafter
Pain		
Secondary infection	Prevent	Monitor daily for sign/symptoms e.g. purulent discharge, fever, swelling)

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects	Prevent/minimize	Duration of therapy

TAKEAWAY

- Burns are categorized by: burning agent, depth, and % total BSA
- Dressings can provide a barrier against contamination, absorb drainage and maintain moisture
 - They are not used for burns on the head, neck and groin
 - Compress dressings include: fine mesh gauze, hydrocolloid dressings, hydrogel dressings, silver-containing dressings, and polyurethane films
- Minor burns can be self-managed with acetaminophen or NSAIDs (avoid ASA), moisturizing lotions, colloidal oatmeal baths, systemic antihistamines, SPF30+, and petrolatum
- Topical antibiotics are routinely used in the initial management of moderate-severe burns
 - Equivalent efficacy amongst topical antibiotics
 - Apply a dressing over the topical antibiotic OR apply a dressing that contains antibiotics
 - Therapeutic options: bacitracin, framycetin, fusidic acid, mupirocin, silver sulfadiazine, chlorhexidine, povidone/iodine
- Moderate-severe burns may require NSAIDs or opioid analgesics for pain relief, and surgery or skin grafts for full thickness burns

CASE STUDY

NK, a 27-year-old male, presents to the walk-in clinic after accidentally spilling hot oil on his right forearm five minutes ago while cooking on a cold winter day. On physical examination, the area appears to be erythematous, moist, blanches with pressure, blistered, and covers approximately 4% of his total body surface area (TBSA). NK reports his pain as 8/10 on the pain scale. His past medical history is significant for allergic rhinitis and asthma. His current medications include loratadine 10 mg PO daily during the summer season, and Symbicort® (budesonide 200 mcg/formoterol 6 mcg) 2 inhalations BID as maintenance therapy and 1-2 inhalations Q4H PRN for the relief of acute symptoms. NK has smoked 10 cigarettes daily for the past 5 years. He does not drink alcohol. NK had an allergic reaction to trimethoprim-sulfamethoxazole that resulted in severe pruritus when he took the antibiotic last year to treat a urinary tract infection.

CASE STUDY

1. What is the most appropriate classification of NK's burn?
 - a) Superficial
 - b) Superficial partial thickness
 - c) Deep partial thickness
 - d) Full thickness
2. Which of the following non-pharmacological measures is most appropriate as initial treatment?
 - a) Run cool water over the affected area for 20 minutes
 - b) Apply butter or ointment to the affected area
 - c) Remove clothing from the burned area
 - d) Apply a cool sterile compress to the affected area
3. Which of the following pharmaceutical agents is LEAST likely to provide benefit for NK's burn injury?
 - a) Silver sulfadiazine 1% applied BID
 - b) Bacitracin ointment applied BID
 - c) Naproxen 500 mg PO BID
 - d) Loratadine 10 mg PO daily
4. What is the expected prognosis of NK's burn injury?
 - a) No scarring and complete symptom resolution
 - b) Discoloration of the skin
 - c) High risk of scarring
 - d) Risk of contracture

SUNBURNS

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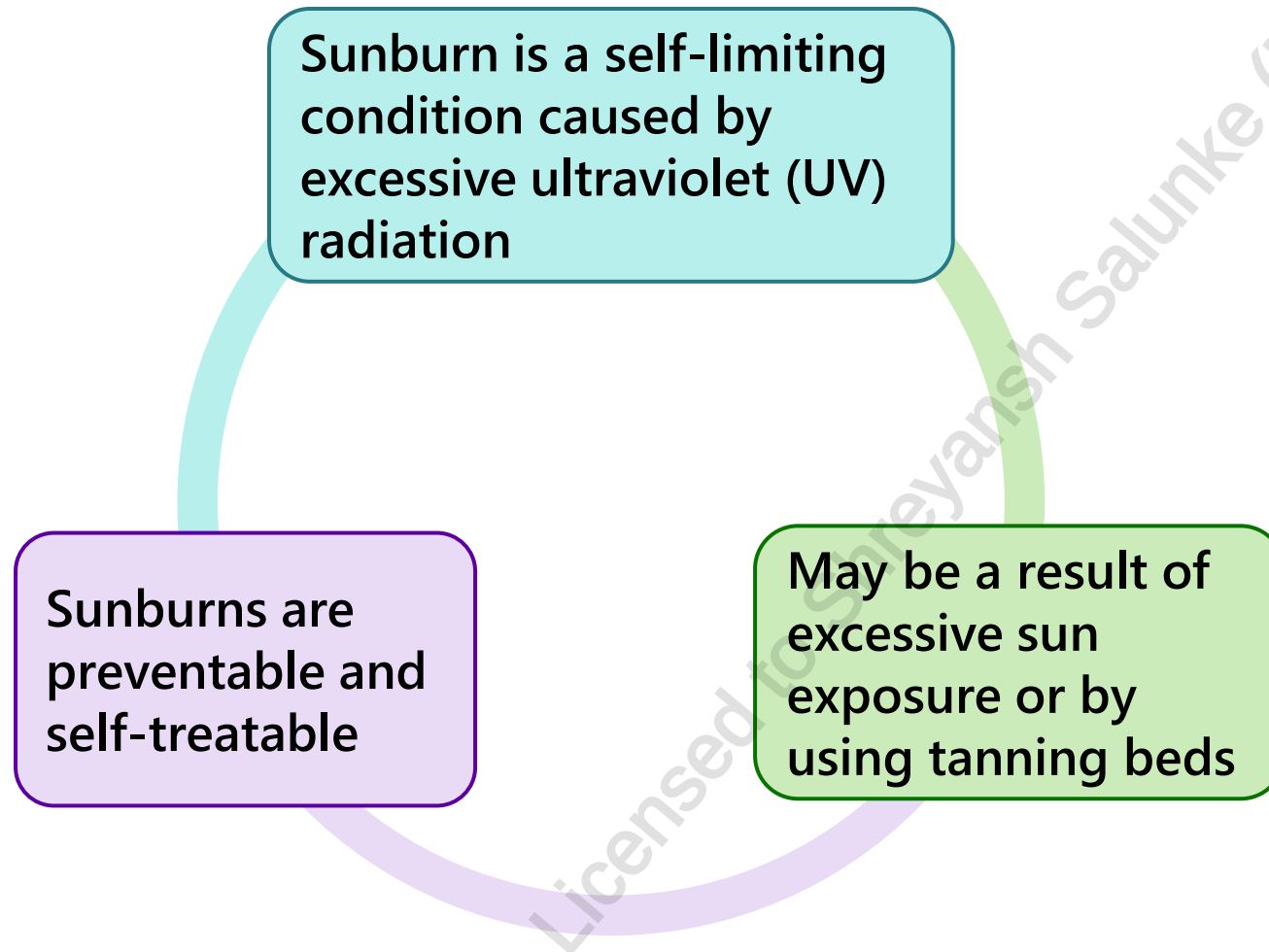


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OBJECTIVES

- 1 Define sunburn
- 2 Discuss the pathophysiology, risk factors, and clinical presentation of sunburns
- 3 Discuss the medication used for prevention and managing symptoms of sunburns
- 4 Outline key patient counseling points about risks of excessive UV exposure

DEFINITIONS



PATHOPHYSIOLOGY

- Ultraviolet rays are categorized by wavelength
 - The shorter the wavelength, the less penetration to the skin
 - However, shorter wavelength UV rays are more harmful
- UVA and UVB cause sunburns, skin cancer, skin aging and wrinkling

UVA

- Long wavelength that penetrates the deeper layers (dermis and subcutaneous fat) of the skin
- Major cause of photosensitive drug reactions
- 95% of UV rays reaching earth

UVB

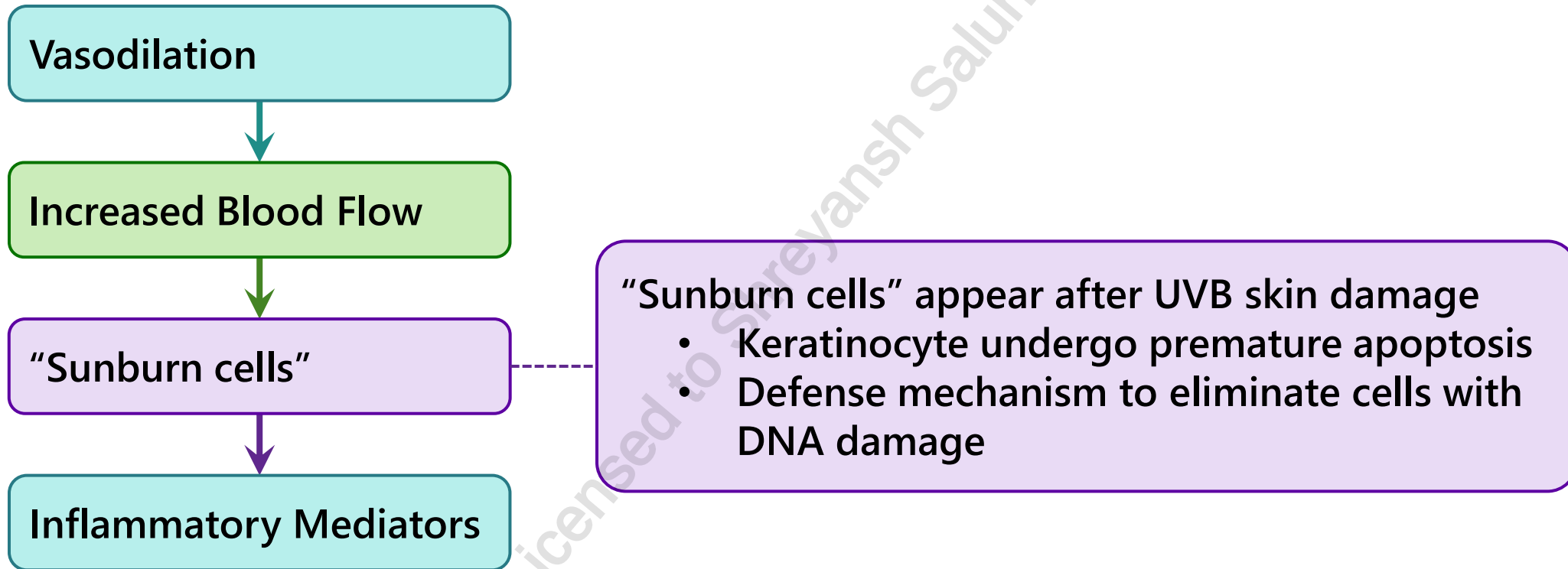
- Medium wavelength that penetrates surface layers (epidermis) of the skin
- Major cause of sunburns
- Ozone layer filters most of the UVB rays (5% of UV rays to reach earth)

UVC

- Short wavelength and the most harmful
- Ozone layer completely filters the UVC rays which prevent the rays from reaching earth

PATHOPHYSIOLOGY OF SUNBURNS

- Ultraviolet rays, mainly UVB rays, penetrate and damage the skin which results in:



CLINICAL PRESENTATION

Signs and Symptoms

- Pain
- Warm/hot skin
- Edema
- Erythema (redness)
- Pruritus
- Peeling skin
- Blisters

Severe cases may also present with

- Fever
- Headache
- Nausea
- Vomiting
- Malaise
- Extensive blistering



EPIDEMIOLOGY

- Individuals with **light skin/eyes** and **light coloured hair** have a higher susceptibility to burning
- Incidence of sunburns is greater among **adolescents and young adults**
- Women use sun protection methods more frequently than men

RISK FACTORS

Sun Exposure

- Beach, outdoor work/sports
- UV index (0-11+)

Tanning Beds

Photosensitive Drugs

Alcohol

- Evidence that alcohol consumption increases chance and size of sunburns

Time of Day

- 11 AM-3 PM → UV rays take a direct path to Earth

Latitude

- UV rays have a shorter distance to travel to reach the equator

Altitude

- UV levels increase 10% with every 1000 m

Ground Reflection

- Snow reflects about 90% of UV radiation
- Sand reflects ~15-30%
- Water reflects ~20%

DRUG RELATED CAUSES

Medications that can cause **phototoxic/photoallergic reactions**:

- **Antimicrobials**: tetracyclines, quinolones, azole antifungals
- **Antimalarials**: quinine, chloroquine
- **Antineoplastics**: 5-FU, vinblastine, paclitaxel, EGF inhibitors
- **Diuretics**: furosemide, HCTZ
- **Psychiatric medications**: chlorpromazine, prochlorperazine
- **NSAIDs**: ibuprofen, diclofenac, naproxen, piroxicam
- **Retinoids**: adapalene, tazarotene, tretinoin
- **Amiodarone, diltiazem, sulfonylureas, coal tar derivatives**



Counsel patients to minimize sun exposure when using these medications

DIAGNOSIS

- Diagnosis is based on **clinical presentation** and **history of exposure to UV rays**
- **Erythema is usually seen 2-6 hours following exposure**, peaks in about a day, and recedes on the third day
- **Sunburn is a self-limiting condition** that usually resolves in 3-7 days
- **Blisters should heal without scarring**
 - May take over a week to heal

Phototoxic Reactions

- Dose-related, sunburn-like reactions that can occur in anyone with sufficient sun and medication exposure

Photoallergic Reactions

- Delayed hypersensitivity reactions after exposure to light that occurs in only small proportion of people

GOALS OF THERAPY

- ✓ Provide symptom relief
- ✓ Educate patient on prevention of sunburns and complications

NON-PHARMACOLOGIC THERAPY

Prevention

- Avoid tanning salons
- Avoid peak hours of sun exposure: 11 AM-3 PM
- Utilize shade when outdoors or use a sun umbrella
- Wear pants, long-sleeve shirts, hats for protection
 - Tightly woven, dark, dry clothing provides greater protection
- Reduce sun exposure when taking photosensitive drugs
- Avoid alcohol when outdoors



Treatment

- Avoid further sun exposure
- Wet compresses for 20 minutes 4-6x/day
- Cool baths
- Replenish fluids
- Keep blisters intact (clean & dress if ruptures)



PHARMACOLOGIC THERAPY – PREVENTION

Sunscreens

- Inorganic/physical sunscreens
 - Photostable filters that reflect and scatter UV & visible light
 - Titanium dioxide, zinc oxide, kaolin, talc, ferric chloride, and melanin
 - Broad spectrum protection with lower risk of irritation or sensitization
 - But its occlusive effects may cause miliaria (heat rash) and folliculitis
 - Classically known to be cosmetically unappealing due to a white film over the skin
- Organic and inorganic ingredients are commonly combined to provide adequate protection against UVA and UVB rays
- Sunscreens should always be used adjunctively as they cannot provide complete protection

PHARMACOLOGIC THERAPY – PREVENTION

Class	Examples	Spectrum	Comments
Cinnamates	Octinoxate	UVB	Octinoxate is the most common Does not adhere well to skin
Salicylates	- Homosalate - Octisalate - Triethanolamine salicylate	UVB	Well tolerated but poorly adheres to the skin
Benzophenones	- Dioxybenzone - Oxybenzone - Sulisobenzene	UVA 2 + UVB	Photostable May cause contact dermatitis Commonly used
Benzotriazole-based	Bisoctrizole	UVB + UVA	Inorganic/organic properties Microfine particles (100-200 nm)
Dibenzoylmethanes	Avobenzone	Broad UVA	+++UVA protection, sensitizing
Para-aminobenzoic acid (PABA) esters	p-aminobenzoic acid	UVB	Adheres well to skin Hypersensitivity (cross-sensitivity with anesthetics/sulfa drugs)

TIPS ON CHOOSING A SUNSCREEN

- **Broad-spectrum** to protect against UVA and UVB rays
 - Products with a critical wavelength of at least 370 nm (blocking 90% of UVA rays) are designated “broad spectrum” by Health Canada
- Sun protection factor (SPF) provides additional protection against UVB rays
 - **Preferably $SPF \geq 30$**
- Most people don't apply enough sunscreen
 - **Aim for coverage of 2 mg/cm^2**
- **SPF protection decreases when used concurrently with DEET**
 - Delay application to maintain protection
- **Water-resistant** sunscreens continue to be recommended by dermatologists
 - Water-resistant for 40 minutes or 80 minutes (refer to product labelling)

How to apply

- Apply generous amounts of sunscreen to all exposed skin
- **Apply 15-30 minutes before exposure to the sun**
- **Reapply every 2 hours or sooner after swimming and/or sweating**



THERAPEUTIC TIPS – PREVENTION

- Patients who have a high susceptibility to sunburns have a greater risk for complications:
 - Skin cancer (especially BCC/SCC)
 - Skin aging and wrinkles
 - Photosensitivity disorders
- Use sunscreen as preventative measure
 - Sunscreen does not guard against all UV rays and does not provide full protection against the sun
 - Decreases the risk of complications (cancer, photoaging)
- Sunscreen is very well tolerated
 - Side effects are rare: hypersensitivity, contact dermatitis

PHARMACOLOGIC THERAPY - TREATMENT

Analgesics

- Acetaminophen used for pain
- NSAIDs used for pain and inflammation

Pramoxine HCl 1%

- Topical anesthetic with low sensitizing potential
- Provides short term relief
- Avoid topical benzocaine because they are sensitizing

Calamine Lotion/Colloidal Oatmeal Bath

- Relieves pain and discomfort

Moisturizers

- Limits dryness and peeling
- Aloe Vera is commonly used, but there is limited evidence of efficacy

Topical Diclofenac Gel (reserved as last resort)

- Apply 6-10 hours after irradiation
- Effective for pain, erythema, and edema
- Do not apply to broken or open skin

PHARMACOLOGIC THERAPY – TREATMENT

Severe Burns

Ruptured blisters

- May require a **topical antibiotic** for prevention or treatment of secondary infections

Hospitalization

- For **severe, large burn area**, signs of **dehydration** or **significant comorbidity**
- **IV fluid replacement**
- **Parenteral analgesics/opioids**
- **Systemic antibiotics (secondary infection)**

THERAPEUTIC TIPS – TREATMENT

Pregnancy/Breastfeeding

- **Physical and chemical sunscreens** are considered safe to use
- **Calamine lotion or colloidal oatmeal** baths for treatment
- **Do not use anti-inflammatory drugs** as treatment during 3rd trimester
 - **Acetaminophen** is safe to use for pain relief

- Do not use topical or systemic corticosteroids
- After sunburn, skin should not be exposed to sun for 1 week

Infants < 6 months old

- Use **non-pharmacologic measures**: create/seek shade and use protective clothing
- Only use **physical sunscreen for small areas**: hands and face

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Erythema	Improve/eliminate	24-48 hours
Sunburn	Heal	
Pain	Improve/alleviate	
Blisters	Heal	> 1 week
Secondary infection	Prevent	Monitor daily for 7 days (especially if blisters are present)

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects (e.g. contact dermatitis)	Prevent	Duration of therapy

TAKEAWAY

- Sunburns can result from excessive sun exposure or by using tanning beds
- Sunburns are preventable and self-limiting, usually resolving in 3-7 days
- Patients using photosensitizing drugs should be counselled on minimizing sun exposure when using these medications
- Sunscreens should be used adjunctively as they cannot provide complete protection
 - Inorganic/physical sunscreen ingredients include: titanium dioxide, zinc oxide, kaolin, talc, ferric chloride, and melanin
 - Organic sunscreen ingredients include: cinnamates, salicylates, benzophenones, benzotriazole-based, dibenzoylmethanes, para-aminobenzoic acid (PABA) esters
 - Organic and inorganic ingredients are commonly combined to provide adequate protection against UVA and UVB rays
 - Sunscreens with $SPF \geq 30$ are recommended
- Pharmacologic management of sunburns includes: acetaminophen/NSAIDs, pramoxine, calamine lotion/colloidal oatmeal bath, moisturizers, topical diclofenac gel
- Severe burns are managed with topical antibiotics for ruptured blisters, and may require hospitalization

CASE STUDY

HJ, a 19-year-old female, presents to the pharmacy with erythema, pruritus, and peeling skin on her upper back and shoulders that is hot to the touch. She does not have any blisters and denies having a fever, headache, nausea, vomiting or malaise. She explains that she spent the day at the beach yesterday and that the temperature outside was 34°C. HJ forgot to take her sunscreen with her but applied a lotion with sun protection factor (SPF) 15 in the morning on her face and body before she left. She spent the first half an hour at the beach in the water, and then she sunbathed for the next 4 hours. Her past medical history is significant for migraine headaches without aura, dysmenorrhea, and acne. Her current medications include sumatriptan 50 mg PO as a single dose at the onset of a migraine attack and a repeat dose if partial relief after ≥ 2 hours (initiated 1 year ago), mefenamic acid 500 mg PO at the onset of bleeding and then 250 mg PO Q6H for 3 days (initiated 1 year ago), doxycycline 100 mg PO daily (initiated 3 months ago), and ethinyl estradiol 30 mcg/desogestrel 0.15 mg PO daily (initiated 6 months ago). HJ also tells you that she is currently in the middle of her monthly cycle. She does not smoke or drink alcohol.

CASE STUDY

1. Which of the following factors is most likely to have increased HJ's risk of photosensitivity?
 - a) Mefenamic acid
 - b) Applying SPF 15 lotion
 - c) Doxycycline
 - d) Ethinyl estradiol/desogestrel
2. Which of the following counselling points is LEAST appropriate to tell HJ about treating her sunburn episode?
 - a) Limit sun exposure to 2 hours a day
 - b) Drink plenty of water to replenish fluids
 - c) Take a cool bath at home
 - d) Apply a wet compress for 20 minutes 2 to 6 times daily
3. Which of the following agents is most appropriate to recommend to HJ to prevent future sunburn reactions?
 - a) Oxybenzone
 - b) Triethanolamine salicylate
 - c) Cinoxate
 - d) Avobenzone
4. Which of the following counselling points is LEAST appropriate to tell HJ regarding sunscreen application?
 - a) Reapply after swimming and/or sweating
 - b) Apply a thin layer all over the face and body
 - c) Apply 15-30 minutes before exposure to the sun
 - d) Recommend a cream or lotion with SPF ≥ 30

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CHANGE LOG

March 2025

- Content reviewed; no changes made
- Slide 14: BSA change from >15% for adults and >10% for children.
Reference added

April 2025

- Content reviewed
- Slide 16: Minor edits
- Slide 33: Updated UV index range to 0-11+
- Slides 33 and 37: Changed peak sun exposure from 10am-4pm to 11am-3pm in line with Canadian references
- Slide 40: Added points on water-resistance and sunscreen reapplication frequency
- Slide 45: Updated references

June 2025

- Slide 3: Added Legend
- Slides 27 and 49: Added Takeaway slides
- Slides 28, 29, 50, 51: Added case study and questions

September 2025

- Slide 27: Added monitoring parameters
- Slide 50: Added monitoring parameters

February 2026

- Slide 44: Removed cinoxate as not available on the Canadian market anymore
- Slides 45 and 51: Updated SPF recommendation
- Slide 47: Updated info on topical diclofenac
- Slide 54: Updated references